
BY 509: Practical OMICs

Course Code	BY 509
Course Name	Practical OMICs
L-T-P-C	0.5-0.5-2-3
Prerequisites	Any MS/MSc/PhD students/BTech students with IC 136 course (Understanding Biotechnology & its Applications) or Consent of Faculty member.
Intended for	B. Tech. 3rd and 4th year, MS/M.Sc/M.Tech., Ph.D.
Distribution	Elective
Approval	50th BoA (Jul 2023)

Course Contents

- **Theory and Tutorials:** The theoretical aspects of different OMICs methods will be covered. This will include introduction to best experimental practices of experiment designing, sample selection, metadata selection, sample and library preparation for respective OMICs analysis. This will be followed by downstream analysis of OMICs data obtained including read preprocessing, and the respective analysis of OMICs data. (10 hours)
- **Practicals:** Mini projects will be assigned to the student involving one or more OMICs experiments. Experimental protocols will be taught and performed utilising the knowledge gained in Module 1 above on assigned case studies. This will be followed by analysis of the data obtained including hands-on exposure of the state-of-the-art software for respective OMICs data analysis, various available resources, and statistical data analysis for the same.(32 hours)